Error trends: standard, ron and en-cot

Oct 30, 2026 · @Shaurya Kochar

64% of wrong answers (19,583 of 30,421) come from three tasks: day of week, next week's dates, and recurrences. A few mistakes show up at every difficulty: counting one day too many, stopping one step short, and forgetting Feb 29.

Data

All 93,150 responses in `responses/`, across all five difficulties and both languages. 30,421 of them are wrong.

Set	What it is	Responses per difficulty	Wrong
standard	direct + CoT, English + Hindi, 150 questions per file	16,200	28,700
ron	same, with `reasoning_effort: low`, 10 questions per file	1,080	422
en-cot	Hindi question, English reasoning, 50 questions per file	1,350	1,299

A script labels every wrong answer with three things:

- how far off it is, in days, minutes or weekdays;
- the kind of mistake: off by one, wrong direction, whole hours off, 12-hour clock, cut off, and so on;
- what the question involves: crossing midnight, a month end or Feb 29.

Comparisons between sets use only the questions they share: the first 10 per file for ron, the first 50 for en-cot.

Where the errors are

Key finding: 64% of wrong answers (19,583 / 30,421) are on three calendar tasks, and Hindi CoT is barely better than Hindi direct (41.0% vs 45.3% wrong).

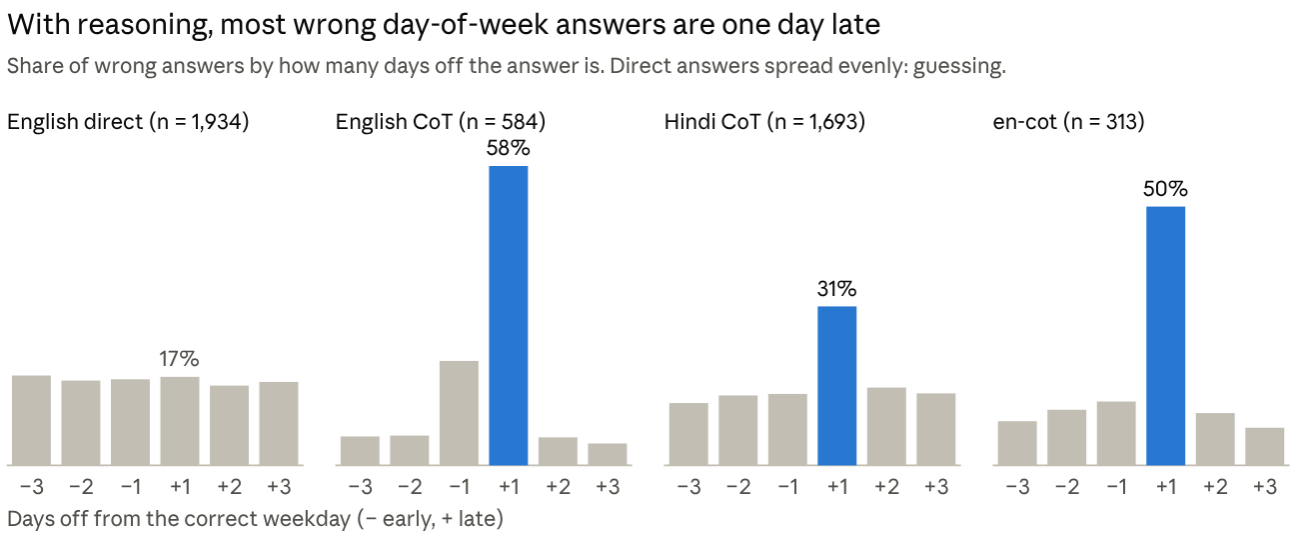
When the model reasons in English (English CoT, en-cot or ron), 78–88% of its wrong answers are on day of week, next week and recurrences. The arithmetic tasks rarely go wrong once it reasons. Direct answers get them wrong too, more so at higher difficulties.

Condition	Wrong overall	Errors on the 3 calendar tasks	Day of week wrong	Next week wrong	Recurrence wrong
standard, English, direct	41.1% (8,322 / 20,250)	61% (5,039 / 8,322)	86% (1,934 / 2,250)	58% (1,306 / 2,250)	80% (1,799 / 2,250)
standard, English, CoT	14.3% (2,903 / 20,250)	85% (2,473 / 2,903)	42% (942 / 2,250)	51% (1,140 / 2,250)	17% (391 / 2,250)
standard, Hindi, direct	45.3% (9,167 / 20,250)	58% (5,332 / 9,167)	85% (1,905 / 2,250)	55% (1,247 / 2,250)	97% (2,180 / 2,250)
standard, Hindi, CoT	41.0% (8,308 / 20,250)	64% (5,299 / 8,308)	79% (1,778 / 2,250)	77% (1,741 / 2,250)	79% (1,780 / 2,250)
ron, English, direct	4.7% (64 / 1,350)	88% (56 / 64)	9% (13 / 150)	17% (26 / 150)	11% (17 / 150)
ron, English, CoT	7.3% (99 / 1,350)	84% (83 / 99)	15% (22 / 150)	20% (30 / 150)	21% (31 / 150)
ron, Hindi, direct	9.1% (123 / 1,350)	78% (96 / 123)	11% (17 / 150)	31% (46 / 150)	22% (33 / 150)
ron, Hindi, CoT	10.1% (136 / 1,350)	85% (116 / 136)	26% (39 / 150)	27% (40 / 150)	25% (37 / 150)
en-cot	19.2% (1,299 / 6,750)	84% (1,089 / 1,299)	54% (404 / 750)	57% (426 / 750)	35% (259 / 750)

Day of week

Key finding: direct answers on day of week are guesses (3,869 wrong answers, 12.7% of all), and with reasoning most errors are exactly one day late.

Direct answers are random guesses: wrong 85% of the time (3,839 / 4,500), which is chance level (85.7%), and spread evenly over the other six days. With English reasoning the errors stop being random: more than half are exactly one day late.



responses/ · wrong day-of-week answers that name a weekday, all difficulties

- **Why one day late:** to count the days from Jan 1, the model adds up the month lengths plus the day of the month. That counts Jan 1 itself, so the total is one too high.
- **Hindi CoT is still mostly guessing:** wrong 79% of the time (1,778 / 2,250), with only a weak one-day-late peak. en-cot, on the same Hindi questions, gets 54% wrong (404 / 750) and shows the clear peak.
- **Dates further out are harder.** English CoT is wrong 30% of the time on short (136 / 450, all 2025 dates) and 50% on very long (223 / 450, dates up to 2033).
- **ron** is wrong only 15% of the time (91 / 600), and many of those responses are cut off at 16k tokens.

Example (English CoT, very_very_long)

> What day of the week is 2029-07-02? Correct: Monday. Model: Tuesday.
>
> "Jan 1, 2024 = Monday … 2029: Jan 1 to July 2 = 31+28+31+30+31+30+2 = 183 … 2010 ÷ 7 = 287 remainder 1. Monday + 1 = Tuesday."

Jan 1 to Jul 2 is 182 days. In another response the model makes the same slip twice, once in its working and once in its check, and takes the matching results as proof it was right.

Example: the same slip in Hindi (Hindi CoT, short)

> 2025-12-12 सप्ताह का कौन सा दिन है? Correct: शुक्रवार (Friday). Model: शनिवार (Saturday).
>
> "346 को 7 से भाग देने पर 49 पूर्ण सप्ताह और 3 दिन शेष … 1 जनवरी 2025, बुधवार था।" (346 ÷ 7 = 49 weeks and 3 days … Jan 1, 2025 was a Wednesday.)

Jan 1 to Dec 12 is 345 days, not 346.

Example: the same slip in en-cot (en-cot, short)

> 2025-03-25 सप्ताह का कौन सा दिन है? Correct: मंगलवार (Tuesday). Model: बुधवार (Wednesday).
>
> "450 days is 64 weeks and 2 days. Starting from Monday, adding 2 days gives Wednesday."

Jan 1, 2024 to Mar 25, 2025 is 449 days.

Recurrences

Key finding: recurrences stop one interval short in 1,698 wrong answers (5.6% of all). Counting backwards is a smaller, Hindi-only pattern (396).

The question asks for the k-th occurrence after today, "without counting today". Two mistakes explain most of the errors we can identify: stopping one interval early, and, on Hindi questions, counting backwards from today.

Condition	Recurrence wrong	One interval short	Counted backwards
standard, English, direct	80% (1,799 / 2,250)	30% (526 / 1,765)	1% (14 / 1,765)
standard, English, CoT	17% (391 / 2,250)	21% (50 / 234)	3% (6 / 234)
standard, Hindi, direct	97% (2,180 / 2,250)	19% (410 / 2,176)	7% (145 / 2,176)
standard, Hindi, CoT	79% (1,780 / 2,250)	35% (612 / 1,758)	9% (164 / 1,758)
en-cot	35% (259 / 750)	48% (99 / 207)	28% (58 / 207)

The last two columns are shares of wrong answers that give a date.

- **One interval short** is the same counting error as in day of week. The model counts today as the first occurrence, so it adds k−1 intervals instead of k.
- **Counting backwards comes from reading Hindi.** en-cot does it in 28% of its errors (58 / 207) even though it reasons in English, against 3% for English CoT (6 / 234). It reads "आज को छोड़कर" (excluding today) as "the occurrence before today".
- **The question wording is ungrammatical:** "the 10 occurrence" in English, "1वीं आवृत्ति" and "हर 1 दिनों" in Hindi. This hurts most on short questions, where k is 1 or 2: en-cot gets 57% of short recurrences wrong (85 / 150) against 25% on medium (38 / 150).

Example: one interval short (Hindi CoT, very_long)

> आज 2031-10-13 है, और मेरे पास हर 25 दिनों पर एक पुनरावृत्ति है। आज को छोड़कर, 10वीं आवृत्ति की तिथि क्या है? Correct: 2032-06-19. Model: 2032-05-25.

The model moves forward 225 days, which is 9 × 25 rather than 10 × 25.

Example: counting backwards (en-cot, short)

> आज 2025-04-10 है, और मेरे पास हर 4 दिनों पर एक पुनरावृत्ति है। आज को छोड़कर, 1वीं आवृत्ति की तिथि क्या है? Correct: 2025-04-14. Model: 2025-04-06.
>
> "Or going forward: 2025-04-14 (4 days after today, first occurrence after today). Given the phrasing 'excluding today,' this suggests we need to find what comes before."

The model finds the right date and then talks itself out of it because of how it reads the Hindi phrase.

Example: one interval short, said outright (English CoT, very_long)

> Today is 2028-10-20, and I have a recurrence every 17 days. Without counting today, what is the date of the 9 occurrence? Correct: 2029-03-22. Model: 2029-03-05.
>
> "If first occurrence is 17 days after today: 9th occurrence = 8 × 17 = 136 days after today."

If the first occurrence is 17 days out, the 9th is 9 × 17 = 153 days out.

Example: counting backwards in Hindi (Hindi CoT, short)

> आज 2025-10-28 है, और मेरे पास हर 2 दिनों पर एक पुनरावृत्ति है। आज को छोड़कर, 1वीं आवृत्ति की तिथि क्या है? Correct: 2025-10-30. Model: 2025-10-26.
>
> "पुनरावृत्ति हर 2 दिनों पर होती है। इसका मतलब है कि पहली बार 2 दिन पहले हुआ था।" (It recurs every 2 days, which means the first time was 2 days ago.)

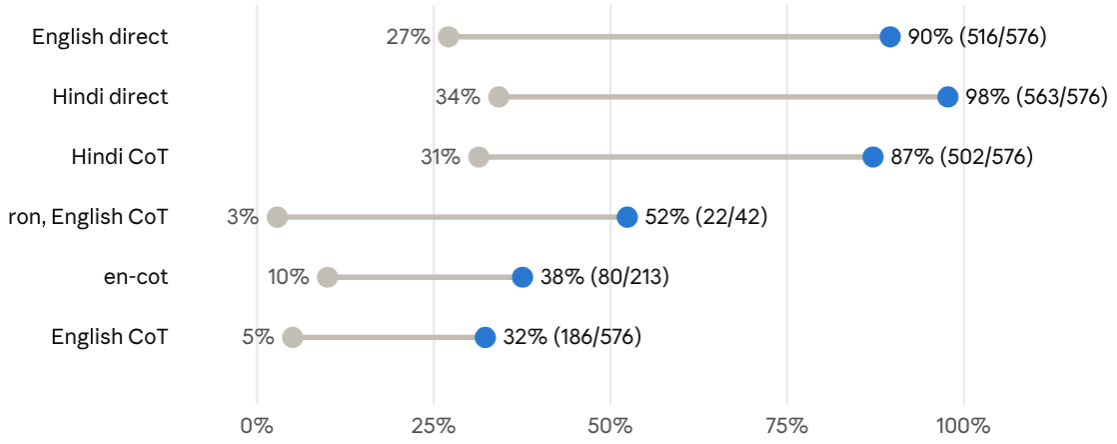
Feb 29 and other boundaries

Key finding: Feb 29 in the span is behind 1,947 wrong answers (6.4% of all) and multiplies the error rate in every condition, ron included.

If the span includes Feb 29, errors go up sharply in every condition, ron included.

Feb 29 in the span multiplies the error rate in every condition

% wrong on date tasks, without (grey) and with (blue) Feb 29 between today and the answer



responses/ · date tasks (addition, subtraction, duration, recurrence), all difficulties pooled · ron has 42 questions with Feb 29 in the span

- **It isn't just a difficulty effect.** Within very_very_long alone, English CoT is wrong 36% of the time with Feb 29 (149 / 411) and 9% without it (124 / 1,389).
- **Month ends and midnight matter much less once the model reasons:**

Trigger	English direct	English CoT	Hindi direct	Hindi CoT	en-cot
Crosses a month end (date tasks)	11% → 47% (2,341 / 4,941)	1% → 11% (555 / 4,941)	17% → 56% (2,772 / 4,941)	15% → 51% (2,523 / 4,941)	8% → 15% (255 / 1,662)
Crosses midnight (time tasks)	20% → 59% (1,408 / 2,370)	2% → 6% (139 / 2,370)	26% → 61% (1,445 / 2,370)	19% → 34% (805 / 2,370)	4% → 7% (52 / 768)

% wrong without → with the trigger; counts are for the "with" side.

Example (English CoT, medium)

> Today is 2028-02-15, and I have a recurrence every 6 days. Without counting today, what is the date of the 4 occurrence? Correct: 2028-03-10. Model: 2028-03-11.
>
> "From 2028-02-27 to 2028-03-05: 28 − 27 = 1 day in February, plus 5 days in March = 6 days ✓"

2028 is a leap year, so that gap is 7 days, and the check just repeats the mistake.

Example: knows it's a leap year, skips Feb 29 anyway (Hindi CoT, medium)

> आज 2028-02-28 है, 6 दिनों में तिथि क्या होगी? Correct: 2028-03-05. Model: 2028-03-06.
>
> "1 दिन बाद: 1 मार्च 2028 … 6 दिन बाद: 6 मार्च 2028 … चूंकि 2028 एक लीप वर्ष है … 28 फरवरी के बाद 29 फरवरी आता है।" (1 day later: March 1 … 6 days later: March 6 … since 2028 is a leap year, Feb 29 comes after Feb 28.)

Example: en-cot (en-cot, medium)

> आज 2028-02-27 है, और मेरे पास हर 4 दिनों पर एक पुनरावृत्ति है। आज को छोड़कर, 3वीं आवृत्ति की तिथि क्या है? Correct: 2028-03-10. Model: 2028-03-11.
>
> "2028-02-27 + 4 days = 2028-03-03"

It should be 2028-03-02. Every later step carries the extra day.

Example: reasoning on (ron, English direct, very_very_long, from the hidden reasoning)

> Today is 2035-10-23, and I have a recurrence every 44 days. Without counting today, what is the date of the 20 occurrence? Correct: 2038-03-21. Model: 2038-03-22.
>
> "Check for possible issues: The date includes February? Wait we didn't cross February 29 or any ambiguous date."

The span does include 2036-02-29. The model even notes earlier that 2036 is a leap year.

Next week's dates

Key finding: with reasoning, next-week answers are mostly one day off. Together with day of week that's 2,461 wrong answers (8.1% of all). Non-week spans are a smaller pattern (548).

The question gives today's date and asks for next week's Monday and Sunday. Each condition goes wrong in its own way.

Condition	Next week wrong	Gave this week	Off by one day	Saturday to Friday	Not 7 days long
standard, English, direct	58% (1,306 / 2,250)	23% (302 / 1,304)	17% (227 / 1,304)	25% (321 / 1,304)	3% (44 / 1,304)
standard, English, CoT	51% (1,140 / 2,250)	3% (29 / 1,046)	**47% (487 / 1,046)**	9% (89 / 1,046)	6% (63 / 1,046)
standard, Hindi, direct	55% (1,247 / 2,250)	12% (149 / 1,243)	26% (319 / 1,243)	20% (247 / 1,243)	5% (56 / 1,243)

Condition	Next week wrong	Gave this week	Off by one day	Saturday to Friday	Not 7 days long
standard, Hindi, CoT	77% (1,741 / 2,250)	5% (86 / 1,720)	23% (392 / 1,720)	10% (166 / 1,720)	**21% (357 / 1,720)**
en-cot	57% (426 / 750)	2% (7 / 410)	**39% (160 / 410)**	8% (34 / 410)	6% (24 / 410)

All columns after the first are shares of wrong answers that give two dates.

- **Direct answers** often give this week instead of next.
- **Reasoning is usually off by one day,** for the same reasons as in day of week: it gets today's weekday wrong by one, or skips a day at a month end.
- **Hindi CoT often returns spans that are not a week at all,** such as Jul 28 to Jul 31. It starts from the wrong weekday and never checks that the span is 7 days long.

Example: not a week (Hindi CoT, long)

> यदि आज 2030-05-15 है, तो अगले सप्ताह की शुरुआत और अंत की तिथियां क्या हैं …? Correct: 2030-05-20 to 2030-05-26. Model: 2030-05-18 to 2030-05-19.
>
> "… 15 मई 2030 शनिवार आता है … अगला सोमवार 18 मई 2030 को होगा … रविवार 19 मई 2030 को होगा (सोमवार से 6 दिन बाद)।"

May 15, 2030 is a Wednesday, not a Saturday. The model also calls May 19 "6 days after Monday", even though it is the day after May 18.

Example: a month end skipped (en-cot, medium)

> यदि आज 2027-05-27 है … Correct: 2027-05-31 to 2027-06-06. Model: 2027-06-01 to 2027-06-07.
>
> "This week ends on Sunday, 2027-05-30. … Next week starts on Monday, 2027-06-01."

Everything is right until the step after May 30, which skips May 31.

Example: this week instead of next (English direct, long)

> If today is 2026-09-13, what are the dates for the start and end of next week, assuming Monday is the first day of the week? Correct: 2026-09-14 to 2026-09-20. Model: 2026-09-07 to 2026-09-13.

Sep 13 is a Sunday, so the model gives the week that ends today.

Example: a 4-day "week" (Hindi CoT, very_long)

> यदि आज 2029-07-24 है … Correct: 2029-07-30 to 2029-08-05. Model: 2029-07-28 to 2029-07-31.
>
> "वर्तमान सप्ताह का रविवार 2029-07-27 होगा। अगला सोमवार … 2029-07-28 होगा। अगला रविवार … 2029-07-31 होगा।" (This week's Sunday is 2029-07-27. Next Monday is 2029-07-28. Next Sunday is 2029-07-31.)

It calls today a Wednesday (it's a Tuesday), then puts "next Sunday" three days after "next Monday".

Time arithmetic

Key finding: 3,760 wrong time answers (12.4% of all) have the minutes right and the hours wrong.

In the time tasks the minutes are usually right and the hours are wrong: 41–65% of wrong time answers are off by a whole number of hours.

Condition	Time tasks wrong	Off by whole hours	Off by exactly 12 hours
standard, English, direct	34% (2,284 / 6,750)	**65% (1,488 / 2,284)**	3% (65 / 2,284)
standard, English, CoT	3% (211 / 6,750)	51% (86 / 169)	15% (25 / 169)
standard, Hindi, direct	38% (2,571 / 6,750)	**57% (1,455 / 2,570)**	3% (81 / 2,570)
standard, Hindi, CoT	24% (1,644 / 6,750)	41% (659 / 1,622)	4% (69 / 1,622)
en-cot	5% (111 / 2,250)	57% (58 / 101)	9% (9 / 101)

The last two columns are shares of wrong answers that give a time.

- **Almost all of these come from direct answers and Hindi CoT** (3,602 of the 3,760). With English reasoning the time tasks are rarely wrong.
- **Hindi CoT gets wrapping around midnight wrong,** for example "24 − 19 = 5, so add 5 hours". Its error rate on time addition stays at 24–27% at every difficulty, so the cause is a fixed method, not bigger numbers.
- **English CoT sometimes switches to the 12-hour clock** (25 / 169 of its time errors) and doesn't convert back.

- **"0 hours" only confuses en-cot** (a small pattern). On questions like "0 घंटे और 4 मिनट" (0 hours and 4 minutes) it is wrong 19% of the time (25 / 129), against 5% on other questions (71 / 1,371). Standard CoT doesn't have this problem.

Example: wrapping around midnight (Hindi CoT, very_very_long)

> अभी 10:46 बजे हैं, 19 घंटे और 41 मिनट बाद समय क्या होगा? Correct: 06:27. Model: 16:27.
>
> "24 - 19 = 5 घंटे। तो, 19 घंटे और 41 मिनट का मतलब है कि हमें 5 घंटे और 41 मिनट जोड़ने हैं।" (24 − 19 = 5 hours, so we add 5 hours 41 minutes.)

Example: 12-hour clock (English CoT, long)

> It is now 07:22, what was the time 15 hours and 31 minutes ago? Correct: 15:51. Model: 03:51.
>
> "6 - 15 = -9 hours. Since I have -9 hours, I need to go back 12 hours to get into the previous day: -9 + 12 = 3 hours."

Example: "0 hours" read as 4 hours (en-cot, short)

> अभी 12:11 बजे हैं, 0 घंटे और 4 मिनट पहले क्या समय था? Correct: 12:07. Model: 08:07.
>
> "I need to find what time it was 4 hours and 4 minutes before 12:11."

Smaller pattern: tens digits and copied numbers in direct answers

Covers 854 tens-digit slips (2.8% of all wrong answers) and 273 answers that reuse a distractor number.

Without reasoning, about a third of wrong date answers are off by exactly 10, 20 or 30 days, and wrong answers reuse a number from the distractor more often than chance. Both mostly disappear once the model reasons in English.

Condition	Date errors of exactly 10, 20 or 30 days	Reuses a distractor number	Chance level
standard, English, direct	35% (338 / 963)	6.2% (66 / 1,065)	3.5% (38 / 1,076)
standard, Hindi, direct	31% (380 / 1,246)	11.2% (141 / 1,255)	4.0% (47 / 1,168)

Condition	Date errors of exactly 10, 20 or 30 days	Reuses a distractor number	Chance level
standard, English, CoT	3% (6 / 213)	0.5% (2 / 364)	0.6% (2 / 325)
standard, Hindi, CoT	9% (126 / 1,347)	4.6% (60 / 1,292)	3.1% (37 / 1,196)
en-cot	4% (4 / 94)	2.0% (4 / 204)	1.3% (2 / 150)

Shares of wrong answers that give a value. Chance level is how often a wrong answer to a question with no distractor happens to share a number with that question's distractor sentences.

- **Tens-digit slips:** the last digit is right and the tens digit is off, as if the answer were written digit by digit without carrying.
- **Copying is worst in Hindi direct,** at 2.8 times chance. The answer tends to land on whatever the distractor mentions: a month end, a named month, a year.

Example: tens digit (English direct, very_very_long)

> Today is 2029-03-06, what is the date going to be in 36 days? Correct: 2029-04-11. Model: 2029-04-01.

Example: the distractor names the answer (English direct, medium)

> I always get confused with months that have 31 days. Today is 2026-12-24, what is the date going to be in 8 days? Correct: 2027-01-01. Model: 2026-12-31.

Example: pulled into the named month (Hindi direct, short)

> अगस्त मेरा पसंदीदा महीना है। आज 2025-07-29 है, 1 दिनों में तिथि क्या होगी? (August is my favourite month. Today is 2025-07-29. What is the date in 1 day?) Correct: 2025-07-30. Model: 2025-08-01.

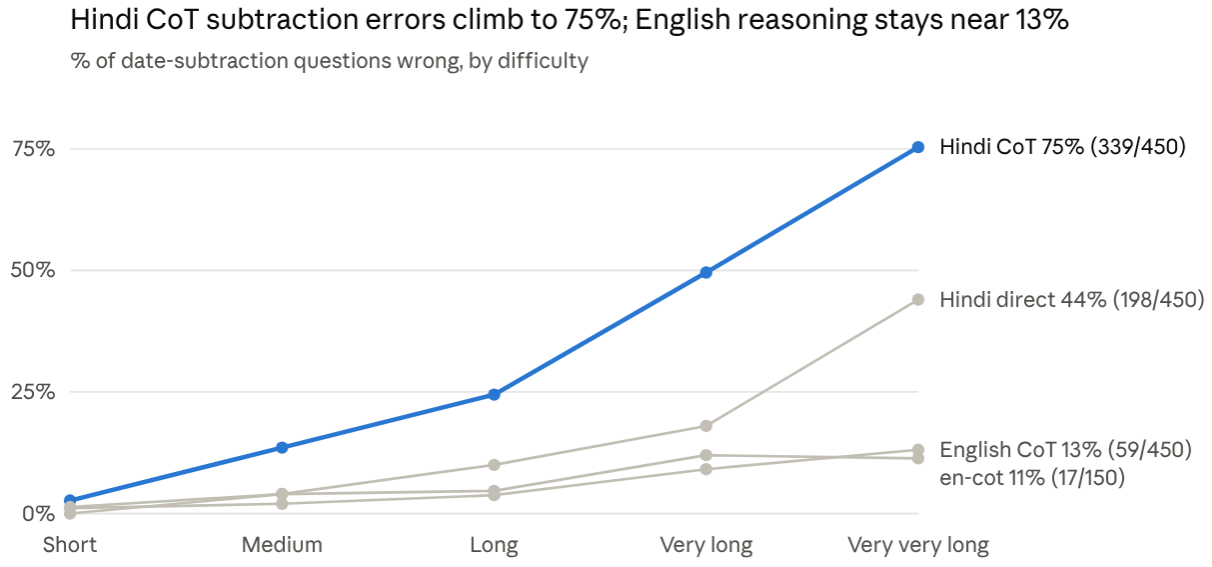
Example: tens digit again (English direct, very_very_long)

> The garden needs new flowers. Today is 2036-04-08, what is the date going to be in 38 days? Correct: 2036-05-16. Model: 2036-05-06.

Smaller pattern: Hindi CoT subtracts by walking forward

Covers 212 wrong answers directly (0.7% of all), but explains why Hindi CoT does worse than Hindi direct on subtraction.

On date subtraction, Hindi CoT gets worse much faster than any other condition. The reason is one habit: it subtracts using the method for adding. It counts forward to the end of the current month, then steps through the next month instead of the previous one.



responses/ · date-subtraction questions, 450 per difficulty for standard (150 for en-cot)

Condition	Answer on or after today	One or two months off
standard, Hindi, CoT	**13% (98 / 738)**	**22% (165 / 738)**
standard, Hindi, direct	5% (16 / 346)	16% (57 / 346)
standard, English, CoT	0% (0 / 129)	4% (5 / 129)
en-cot	0% (0 / 45)	4% (2 / 45)

Shares of wrong subtraction answers. "One or two months off" means 27–32 or 58–62 days off.

en-cot reads the same Hindi questions and doesn't do this, so it comes from reasoning in Hindi, not from reading it. All five very_very_long errors we sampled used this method.

Example (Hindi CoT, very_very_long)

> आज 2031-02-26 है, 54 दिन पहले क्या तारीख थी? (Today is 2031-02-26. What was the date 54 days ago?) Correct: 2031-01-03. Model: 2031-03-08.
>
> "26 फरवरी से 28 फरवरी तक कुल 3 दिन … मार्च में 31 दिन होते हैं … 28 मार्च से 20 दिन पीछे जाने पर 8 मार्च आता है।" (Feb 26 to Feb 28 is 3 days … March has 31 days … 20 days back from March 28 is March 8.)

The answer is ten days after today.

Example: counts to the month end, then back (Hindi CoT, very_very_long)

> आज 2027-05-09 है, 52 दिन पहले क्या तारीख थी? Correct: 2027-03-18. Model: 2027-05-01.
>
> "9 मई से 31 मई तक कुल 22 दिन होते हैं … 52 में से 22 दिन कम हो जाएंगे … 30 अप्रैल से ठीक 30 दिन पहले की तारीख 1 मई होती है।" (May 9 to May 31 is 22 days … take 22 from 52 … exactly 30 days before April 30 is May 1.)

Example: takes off the whole month (Hindi CoT, very_very_long)

> आज 2036-07-14 है, 52 दिन पहले क्या तारीख थी? Correct: 2036-05-23. Model: 2036-06-09.
>
> "जुलाई महीने में 31 दिन होते हैं … 52 में से 31 दिन कम हो जाते हैं, जिससे 21 दिन बचते हैं … जून में 21 दिन पीछे … 9 दिन पर पहुंच जाते हैं।" (July has 31 days … 52 − 31 leaves 21 … go back 21 days in June … we reach the 9th.)

Today is July 14, so only 14 days of July come off, not 31.

Example: goes back a month but keeps the month (Hindi CoT, very_very_long)

> आज 2027-10-01 है, 38 दिन पहले क्या तारीख थी? Correct: 2027-08-24. Model: 2027-09-24.
>
> "पहले अक्टूबर के 31 दिन पीछे जाएंगे, जिससे हम … 2027-09-01 पर पहुंच जाएंगे … 7 दिन और पीछे जाने पर हम 24 तारीख पर पहुंच जाएंगे।" (First go back October's 31 days, which brings us to 2027-09-01 … going 7 more days back we reach the 24th.)

31 days before Oct 1 is Aug 31, and the model also never moves the answer out of September.

Distractors and cut-off responses

Key finding: 1,289 wrong answers (4.2% of all) are responses cut off before answering, including 287 of ron's 422 (68%). The distractor effect itself is small.

A similar distractor (one about dates or times) adds 2–4 points of error when the model reasons, and almost none for direct answers. Many errors with reasoning aren't wrong answers at all: the response runs out of tokens while the model is still re-checking.

Condition	No distractor	Dissimilar	Similar	Wrong answers that were cut off
standard, English, direct	40.0% (2,698 / 6,750)	41.9% (2,830 / 6,750)	41.4% (2,794 / 6,750)	0% (29 / 8,322)
standard, English, CoT	13.7% (928 / 6,750)	13.3% (896 / 6,750)	16.0% (1,079 / 6,750)	22% (651 / 2,903)
standard, Hindi, CoT	39.7% (2,678 / 6,750)	41.1% (2,777 / 6,750)	42.3% (2,853 / 6,750)	2% (146 / 8,308)
ron, English, CoT	5.3% (24 / 450)	7.8% (35 / 450)	8.9% (40 / 450)	**85% (84 / 99)**
ron, Hindi, CoT	10.7% (48 / 450)	9.1% (41 / 450)	10.4% (47 / 450)	**81% (110 / 136)**
en-cot	17.8% (400 / 2,250)	18.2% (409 / 2,250)	21.8% (490 / 2,250)	13% (174 / 1,299)

The first three columns are % wrong.

- **A similar distractor can steer the reasoning.** The model starts discussing the distractor's topic and sometimes gets that topic wrong too. 130 wrong time answers with reasoning discuss the distractor, 105 of them from Hindi CoT.
- **Cut-offs are mostly on the calendar tasks.** Of English CoT's 651 cut-off responses, 358 are day of week, 156 recurrences and 89 next week. Most are loops of re-checking that start after the model notices its own one-day error.
- **ron's errors are mostly cut-offs, not wrong answers.** Aditya's notes in `experiments/2026-10-25_test-reasoning-on/ANALYSIS.txt` show the same loops in the hidden reasoning.

Example: the distractor changes a fact (Hindi CoT, long)

> मुझे 31 दिनों वाले महीनों में भ्रम होता है। आज 2026-09-02 है, 12 दिन पहले क्या तारीख थी? Correct: 2026-08-21. Model: 2026-08-31.
>
> "31 दिनों वाले महीने अप्रैल, जून, सितंबर और नवंबर होते हैं।" (The months with 31 days are April, June, September and November.)

Those are the 30-day months, and the answer lands on the month end the distractor talks about.

Example: a re-checking loop (English CoT, short; cut off at 4,096 tokens)

> What day of the week is 2025-08-21? Correct: Thursday. No answer.
>
> "Day 233: (233-1) mod 7 = 1. Wednesday + 1 = Thursday. But month-by-month gave Friday. There's a 1-day discrepancy. Let me recheck July 31 …"

The model had the right answer and lost it while hunting for its own off-by-one error.

Example: the distractor taken literally (Hindi CoT, short)

> मेरी घड़ी धीमी चलती है। अभी 21:25 बजे हैं, 2 घंटे और 3 मिनट बाद समय क्या होगा? (My watch runs slow. It is 21:25. What time will it be in 2 hours 3 minutes?) Correct: 23:28. Model: 00:00.
>
> "… 23:25 + 3 मिनट = 23:28 … लेकिन … हम 23:28 नहीं निकाल सकते क्योंकि मेरी घड़ी धीमी है।" (… 23:28 … but we can't say 23:28 because my watch is slow.)

It works out the right answer, then throws it away because of the distractor. Hindi CoT answers "00:00" 19 times with a similar distractor, against 5 times with none.

Example: the distractor leads to a cut-off (English CoT, very_very_long)

> My watch always runs a bit slow. It is now 13:44, what was the time 59 hours and 37 minutes ago? Correct: 02:07. No answer.
>
> "Could 'it is now 13:44' refer to the watch? If so, and the watch is slow, then actual time is later. But then we can't find the actual time 59 hours and 37 minutes ago without knowing how slow."

It mentions 02:07 but keeps arguing about the watch until it runs out of tokens. 130 wrong time answers with reasoning bring up the distractor's topic like this, 105 of them from Hindi CoT.

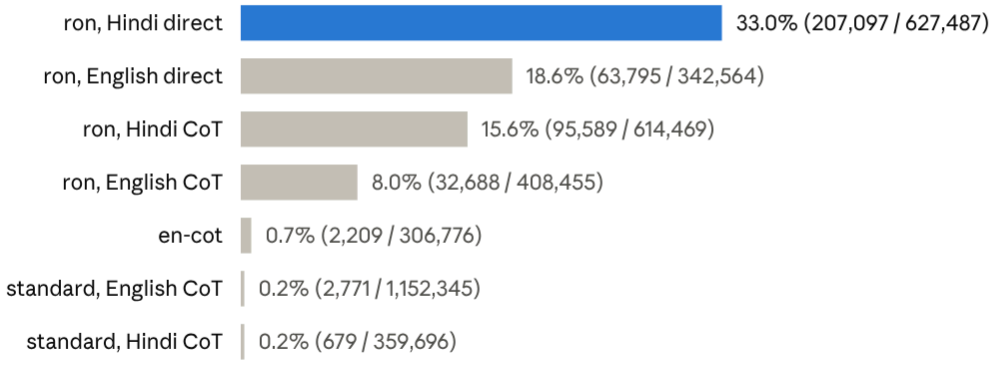
What the parsed reasoning shows

Aditya's parser (`parser.py`) turns each reasoning trace into a sequence of labelled steps: `PLAN`, `ADD`, `SUB`, `LEAP`, `CHECK`, `REVISE` ("wait, actually…"), `IGNORE` (talking about the distractor), `FORMAT` (arguing about output format or language), and so on. We parsed every response that has reasoning: standard CoT, en-cot, and ron's hidden reasoning, 55,167 traces in all. The parser labels 96–97% of sentences in standard CoT and en-cot, and 82–89% in ron.

Key finding: format deliberation fills a third of ron's Hindi direct reasoning (207,097 of 627,487 steps), but it doesn't make answers wrong.

Format deliberation fills a third of ron's Hindi direct reasoning, and almost none of visible CoT

Share of parsed reasoning steps that argue about output format or language (FORMAT)



parsed_traces.csv · 55,167 parsed traces (standard CoT, en-cot, ron hidden reasoning) · steps labelled by parser.py

- **It comes from the prompt's rules.** The Hindi direct prompt asks for "only the final answer", the whole response in Hindi, the word "Answer:" in English and Western digits, all at once. ron's hidden reasoning argues about these instead of the arithmetic. Visible CoT almost never does (0.2–0.7% of steps).
- **It's worst where the maths is easiest:** 39% of ron's steps on date addition and 34–38% on the time tasks, against 9–10% on day of week and next week. It falls from 27% of steps on short to 15% on very_very_long.
- **It doesn't predict errors.** Right ron Hindi direct traces have more FORMAT steps than wrong ones (157 vs 118 on average). It wastes tokens rather than changing answers.

Key finding: long re-checking loops mark wrong and cut-off answers.

- On every task, English CoT's wrong traces have 1.5–6 times more `CHECK` and `REVISE` steps than its right ones. For example, time durations average 12.4 against 2.0, and recurrences 19.3 against 5.0.
- Cut-off English CoT traces average 42 re-checking steps against 8 for wrong traces that did finish. 869 of English CoT's 2,903 wrong traces (30%) have 20 or more.

Key finding: Hindi CoT almost never checks its work. Only 744 of 20,250 Hindi CoT traces (3.7%) contain a `CHECK` or `REVISE` step, against 17,820 / 20,250 (88%) for English CoT and 5,132 / 6,750 (76%) for en-cot. Hindi CoT traces are also much shorter (median 13 steps against 25 for English CoT). This fits it being barely better than answering directly.

- **Feb 29 errors aren't from forgetting leap years.** 185 of 190 wrong English CoT answers on Feb 29 questions mention the leap year (`LEAP`), as do 78 / 83 for en-cot and 416 / 508 for Hindi CoT. The model knows it's a leap year and still counts wrong.

- **Talking about the distractor doesn't predict errors.** Hindi CoT discusses the distractor (`IGNORE`) in 4,079 of 20,250 traces (20%), English CoT in 1,428 (7%). With a similar distractor, it happens equally often in right and wrong Hindi CoT traces (30% vs 30%).
- **What the parser can't show:** Hindi CoT's forward-walking subtraction. Counting forward to the month end is written as a subtraction ("31 − 9 = 22"), so it gets labelled `SUB` like a correct step.

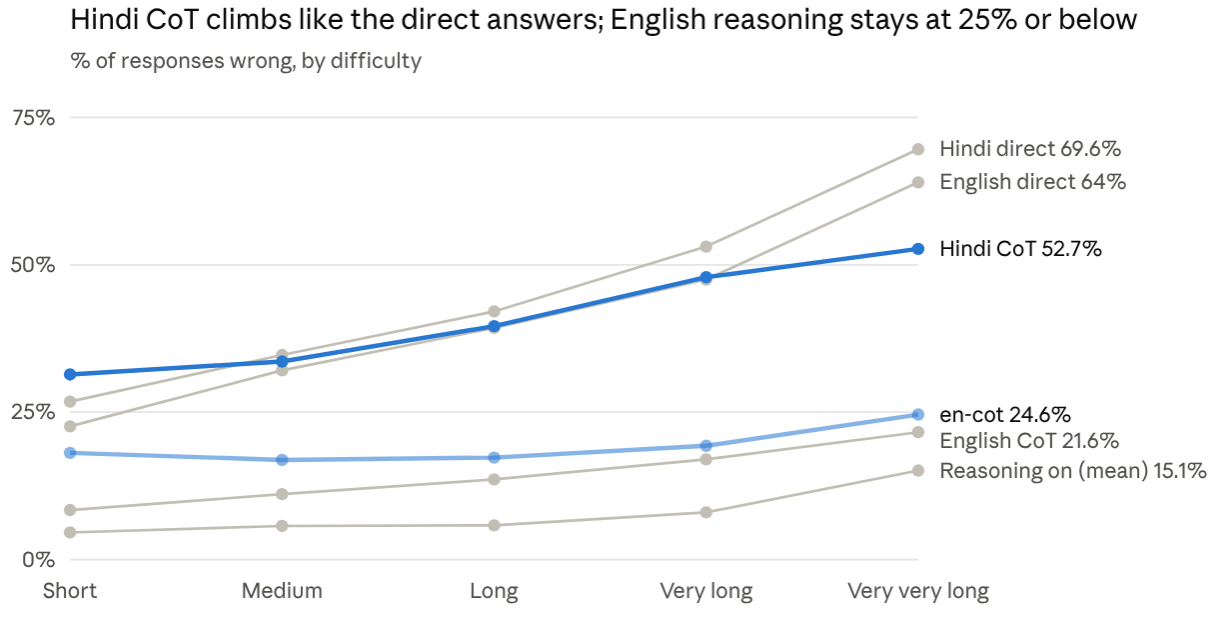
Example: 5,170 words to add one day (ron, Hindi direct, short; answered correctly)

> संग्रहालय नौ बजे खुलता है। आज 2025-04-06 है, 1 दिनों में तिथि क्या होगी? (The museum opens at nine. Today is 2025-04-06. What is the date in 1 day?) Correct: 2025-04-07. Model: 2025-04-07.
>
> "Entire response in Hindi. … Keep the word 'Answer:' in English, use Western digits, and put nothing else on that line. … That's not in Hindi. … But if I have no text outside the last line, then there's no Hindi text."

361 of the trace's 630 parsed steps are `FORMAT` . Other ron Hindi direct traces go the same way: "Is '31 जुलाई 2025' entirely in Hindi?", "Give only the final answer. This is absolute."

By difficulty

Each level pushes dates further out (only 2025 on short, up to 2036 on very_very_long) and doubles the offsets, intervals and durations. Direct answers and Hindi CoT get steadily worse. English CoT and en-cot stay fairly flat until very_very_long.



responses/ · 93,150 responses · reasoning on = mean of its four language × condition cells

Short

2025 dates, offsets of 1–4, durations up to 60 minutes, 1st or 2nd occurrence.

- The arithmetic is almost always right. The errors are on day of week, next week and recurrences.
- Recurrences are harder than on medium: en-cot gets 57% wrong (85 / 150) against 25% (38 / 150), and Hindi CoT 72% (323 / 450) against 57% (256 / 450). With k = 1 or 2, misreading "excluding today" decides the answer: 55 of en-cot's 80 wrong dates here count backwards.
- Next week has the biggest language gap: Hindi CoT is wrong 60% of the time (268 / 450), English CoT 28% (126 / 450).

Medium

2025–2028, offsets of 4–8, durations of 1–2 hours, 2nd–4th occurrence.

- Direct answers on time durations jump from 5% (24 / 450) to 25% wrong (114 / 450) in English and from 10% (47 / 450) to 44% (198 / 450) in Hindi, because durations now cross hours and need borrowing.

Long

2025–2030, offsets of 8–16, durations of 2–4 hours, 4th–8th occurrence.

- Hindi CoT starts walking forward on subtraction: 24% wrong (110 / 450) against 4% for English CoT (17 / 450).
- Next week is wrong more than half the time in every standard condition, from 54% for English CoT (244 / 450) to 83% for Hindi CoT (375 / 450).

Very long

2025–2033, offsets of 16–32, durations of 4–8 hours, 8th–16th occurrence.

- Direct time answers are wrong 40–71% of the time, from 180 / 450 (Hindi, addition) to 321 / 450 (Hindi, durations), mostly off by whole hours.
- English CoT on day of week peaks at 50% wrong (223 / 450).

Very very long

2025–2036, offsets of 32–64, durations of 8–16 hours, 16th–32nd occurrence.

- Direct answers are wrong 64% (2,593 / 4,050) in English and 70% (2,818 / 4,050) in Hindi.

- Recurrences go wrong even with reasoning: English CoT 38% (169 / 450), en-cot 45% (67 / 150), ron 62% (75 / 120, mostly cut off).

Standard vs ron vs en-cot on the same questions

On the first 10 questions per file, which all three sets share, Hindi errors fall from 40% (Hindi CoT) to 18% with en-cot and 10% with ron. Each set leaves a different kind of error behind.

Condition	Wrong	Wrong answers that were cut off	What the remaining errors mostly are
standard, Hindi, direct	42.1% (569 / 1,350)	0% (0 / 569)	Day-of-week guesses, copied distractor numbers, tens-digit slips
standard, Hindi, CoT	40.1% (542 / 1,350)	2% (11 / 542)	Subtracting by walking forward, "weeks" that aren't 7 days, midnight wrap-around mistakes
en-cot	18.1% (245 / 1,350)	13% (32 / 245)	Recurrences counted backwards, one day late, "0 hours" misread
ron, Hindi, direct	9.1% (123 / 1,350)	46% (57 / 123)	Cut off at 16k tokens, Feb 29
ron, Hindi, CoT	10.1% (136 / 1,350)	81% (110 / 136)	Cut off at 16k tokens
standard, English, direct	36.9% (498 / 1,350)	0% (0 / 498)	Day-of-week guesses, this week instead of next, tens-digit slips
standard, English, CoT	12.7% (172 / 1,350)	20% (35 / 172)	One day late, 12-hour clock, re-checking loops
ron, English, direct	4.7% (64 / 1,350)	56% (36 / 64)	Cut off at 16k tokens, Feb 29
ron, English, CoT	7.3% (99 / 1,350)	85% (84 / 99)	Cut off at 16k tokens

- **en-cot fixes the errors that come from reasoning in Hindi** (walking forward, non-week spans) but keeps the ones that come from reading Hindi (counting backwards, "0 hours").

- **ron fixes most reasoning mistakes.** What's left is mostly running out of tokens, plus Feb 29.
- **The one-day-late error is the top day-of-week error for both English CoT (341 / 584) and en-cot (158 / 313),** so it looks like a habit of the model, not a language effect.

Caveats

- ron and en-cot are small: once split by task and difficulty, their cells hold 30–150 responses, so ignore differences of a few points there.
- The labels come from a script. "Other" is still 66% of standard errors, so the labels don't explain every error. The examples were checked by hand; the percentages weren't.
- Cut-off responses count as wrong, which matters most for ron.
- Scoring uses `metrics.py` on `main`. Rahul's newer parser fix may shift the Hindi numbers slightly.

Code and tables: `experiments/2026-10-30_error-trends-shaurya/`. `build_table.py` labels every response, `patterns.py` makes the numbers, and `examples.py` saves every quoted example in full.

`counts.py` produces every n / N count in this doc and the Key findings ranking (`outputs/counts.json`). 3,711 wrong recurrence answers (12.2% of all) don't fit any of our labels, so recurrences have more going on than the two causes above.

`parse_traces.py` runs `parser.py` over every trace (`outputs/parsed_traces.csv`; it needs Python 3.11+), and `parsed_patterns.py` makes the parsed-reasoning numbers (`outputs/parsed_numbers.json`).